In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

In [2]:
```
dataset = pd.read_csv('50_Startups.csv',header=0)
```

In [3]:
```
dataset.head()
```

Out[3]:

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

In [4]:
```
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   R&D Spend        50 non-null     float64
 1   Administration   50 non-null     float64
 2   Marketing Spend  50 non-null     float64
 3   State            50 non-null     object 
 4   Profit           50 non-null     float64
dtypes: float64(4), object(1)
memory usage: 2.1+ KB
```

In [5]:
```
dataset.describe()
```

Out[5]:

	R&D Spend	Administration	Marketing Spend	Profit
count	50.000000	50.000000	50.000000	50.000000
mean	73721.615600	121344.639600	211025.097800	112012.639200
std	45902.256482	28017.802755	122290.310726	40306.180338
min	0.000000	51283.140000	0.000000	14681.400000
25%	39936.370000	103730.875000	129300.132500	90138.902500
50%	73051.080000	122699.795000	212716.240000	107978.190000
75%	101602.800000	144842.180000	299469.085000	139765.977500
max	165349.200000	182645.560000	471784.100000	192261.830000

In [6]:
```
dataset = pd.get_dummies(dataset)
```

In [7]:
```
dataset.columns
```

Out[7]:
```
Index(['R&D Spend', 'Administration', 'Marketing Spend', 'Profit',
       'State_California', 'State_Florida', 'State_New York'],
      dtype='object')
```

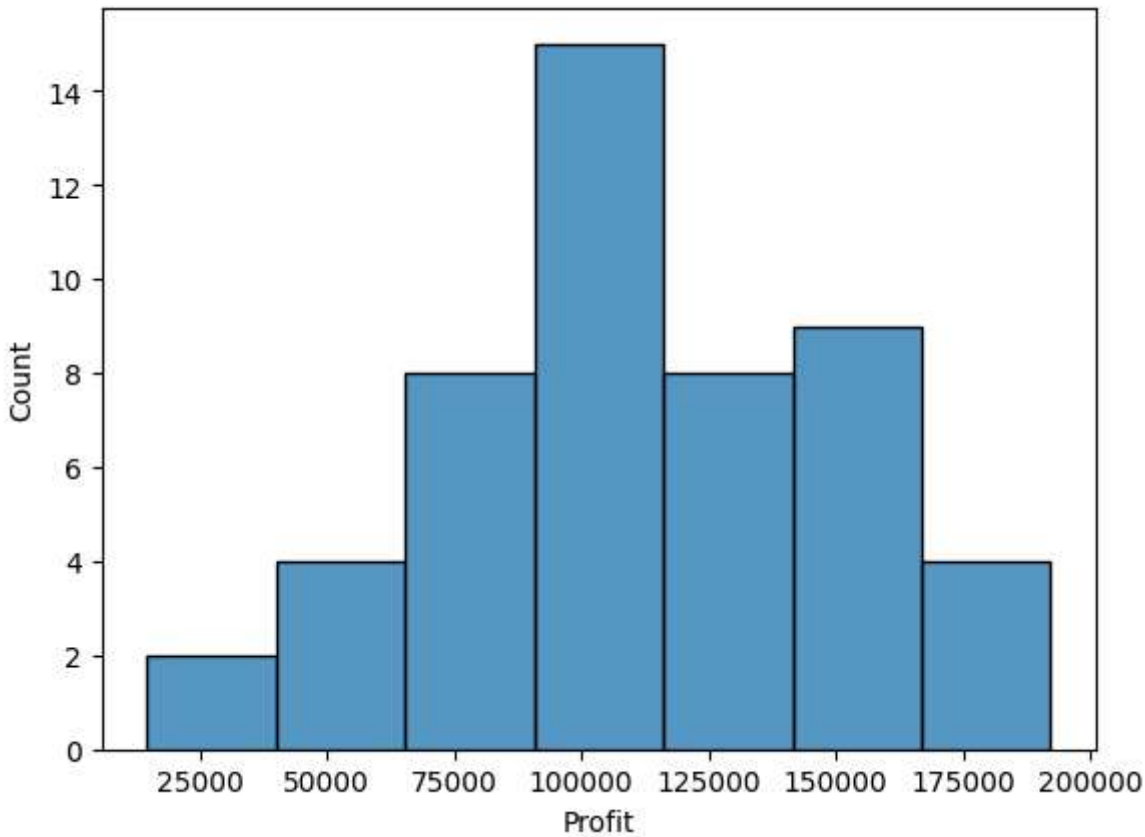
In [8]:
```
dataset.head()
```

Out[8]:

	R&D Spend	Administration	Marketing Spend	Profit	State_California	State_Florida	State_N Y
0	165349.20	136897.80	471784.10	192261.83	False	False	T
1	162597.70	151377.59	443898.53	191792.06	True	False	Fa
2	153441.51	101145.55	407934.54	191050.39	False	True	Fa
3	144372.41	118671.85	383199.62	182901.99	False	False	T
4	142107.34	91391.77	366168.42	166187.94	False	True	Fa

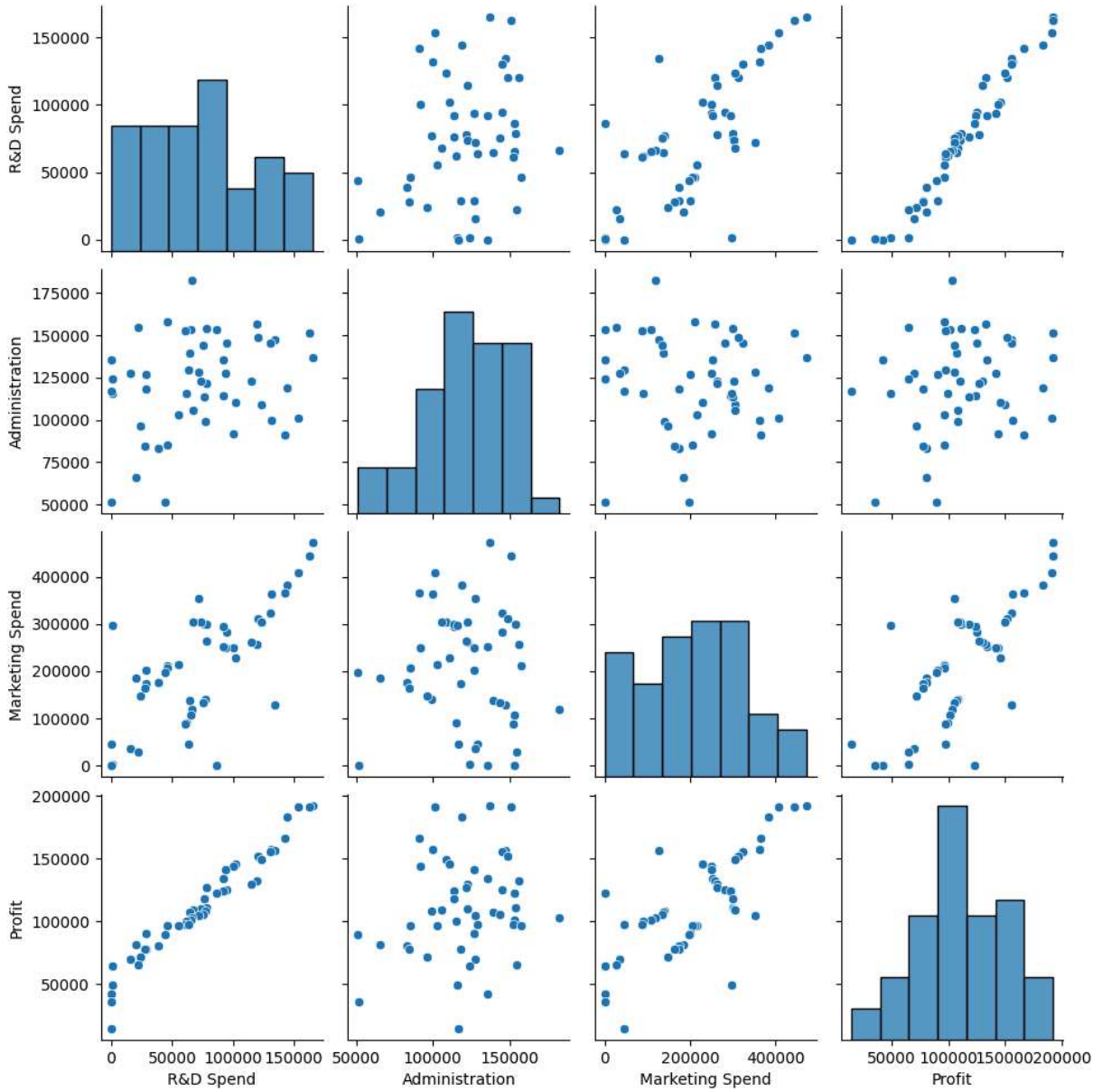
In [9]:
```
sns.histplot(dataset['Profit'])
```

Out[9]:
```
<Axes: xlabel='Profit', ylabel='Count'>
```



In [10]:
```
sns.pairplot(dataset[['R&D Spend', 'Administration', "Marketing Spend", "Profit"]])
```

Out[10]:
```
<seaborn.axisgrid.PairGrid at 0x219cc656660>
```



In [11]:
```
sns.heatmap(dataset[['R&D Spend', 'Administration', "Marketing Spend", "Profit"]].c
```

Out[11]: <Axes: >

In [12]:
```
X = dataset[['R&D Spend', 'Administration', 'Marketing Spend', 'State_California',
y = dataset['Profit']
```

In [13]:
```
from sklearn.model_selection import train_test_split
```

In [14]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=99
)
```

In [15]:
```
from sklearn.linear_model import LinearRegression
```

In [16]:
```
lm = LinearRegression()
```

In [17]:
```
lm.fit(X_train,y_train)
```

Out[17]:
```
▾ LinearRegression
▸ Parameters
```

In [18]:
```
#Intercepto
print(lm.intercept_)
```

```
47169.907977688126
```

In [19]:
```
#Coeficientes
coeff_df = pd.DataFrame(lm.coef_,X.columns,columns=['Coefficient'])
coeff_df
```

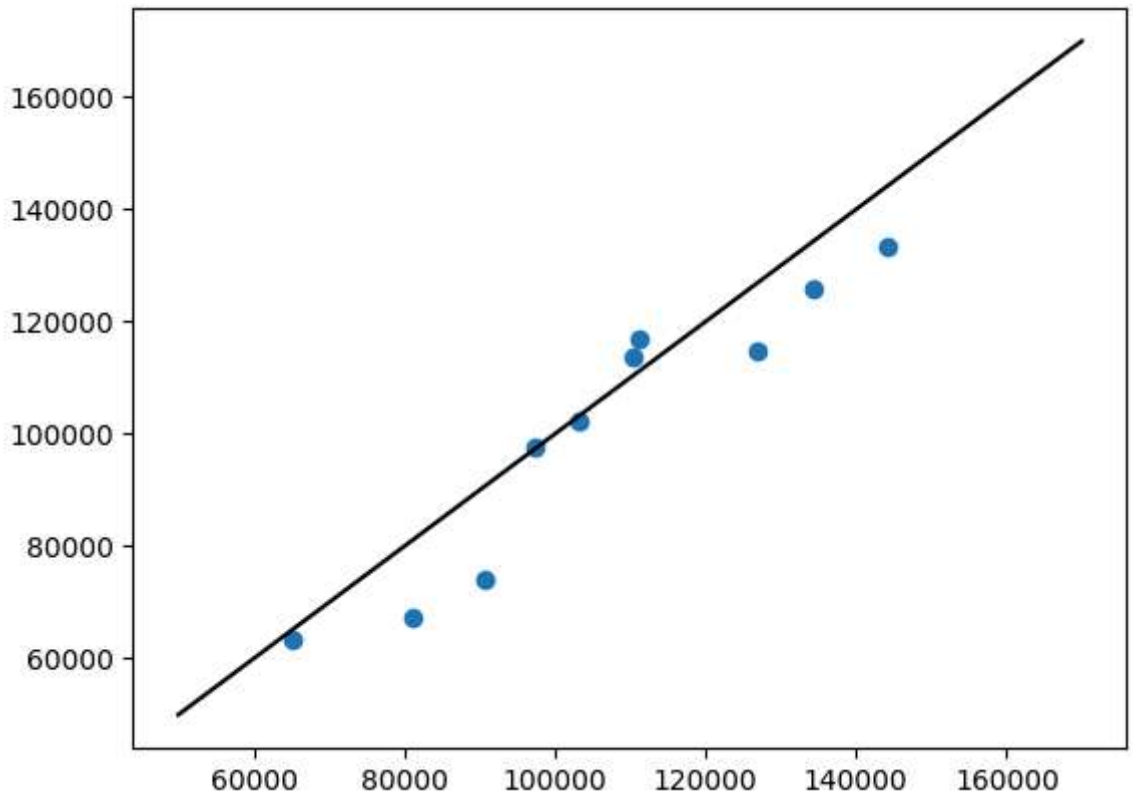
Out[19]:

	Coefficient
R&D Spend	0.820135
Administration	-0.009123
Marketing Spend	0.023009
State_California	-1352.011873
State_Florida	-74.732930

In [20]:
```
predictions = lm.predict(X_test)
```

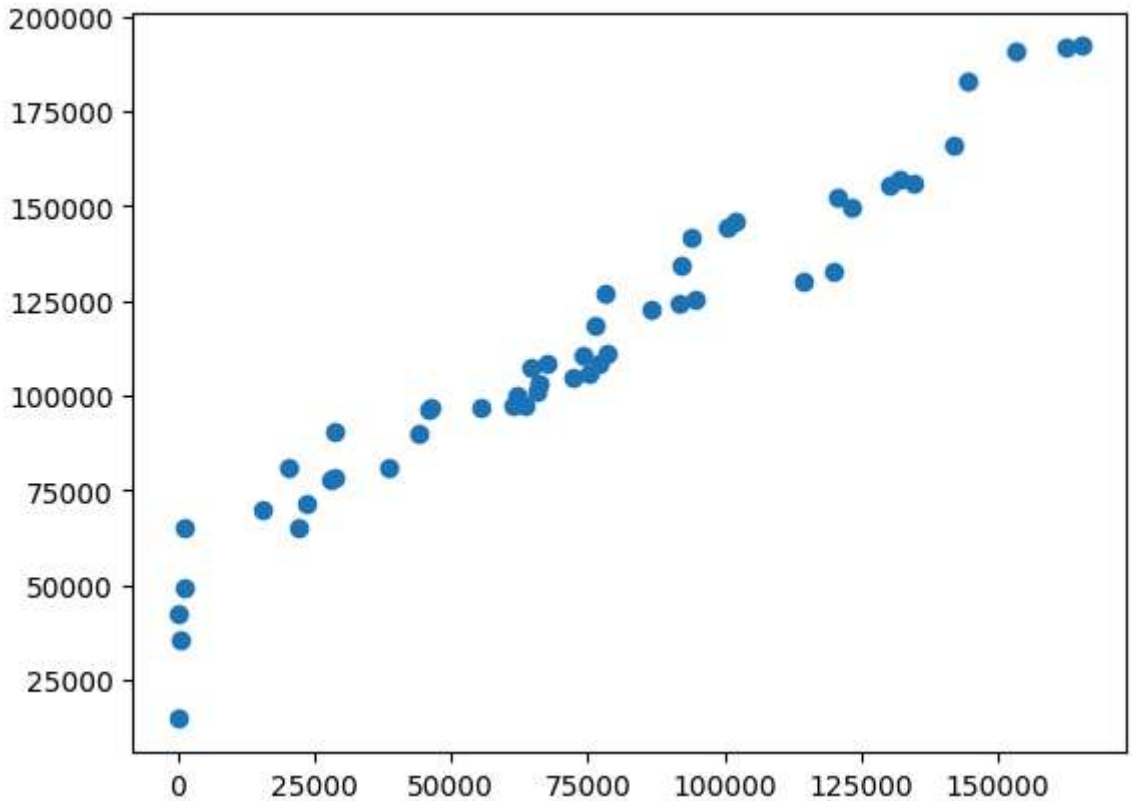
In [21]:
```
plt.scatter(y_test, predictions)
plt.plot([50000,170000], [50000,170000], color = 'black')
```

Out[21]:
```
[<matplotlib.lines.Line2D at 0x219cede42d0>]
```



In [22]:
```
plt.scatter(dataset['R&D Spend'],dataset['Profit'])
```

Out[22]:
```
<matplotlib.collections.PathCollection at 0x219cee4e0d0>
```

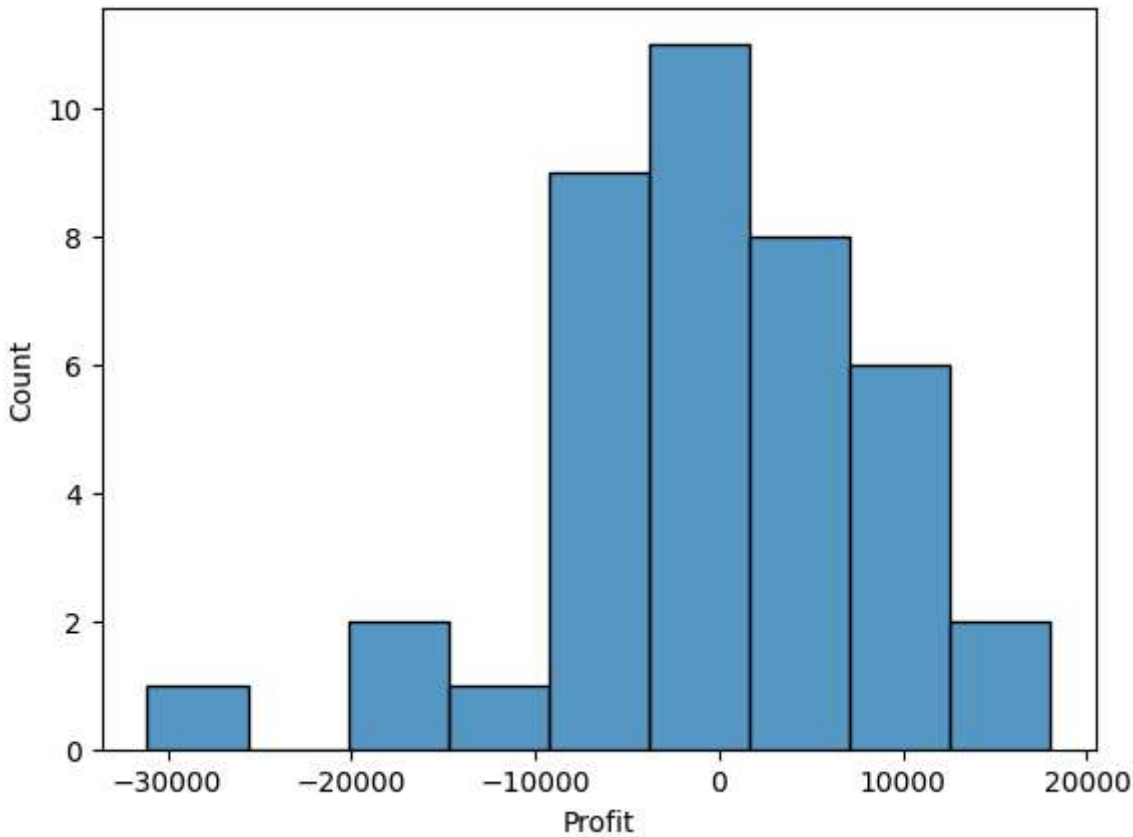


In [23]:
```
np.corrcoef(y_test,predictions)[0][1]
```

Out[23]:
```
np.float64(0.9501983359183575)
```

In [24]:
```
predictions_train = lm.predict(X_train)
```

In [25]:
```
sns.histplot((y_train-predictions_train),bins=9);
```



In [26]:
```
from sklearn import metrics
```

In [27]:
```
print('MAE', metrics.mean_absolute_error(y_test,predictions))
print('MSE',metrics.mean_squared_error(y_test,predictions))
print('RMSE:', np.sqrt(metrics.mean_squared_error(y_test, predictions)))
```

```
MAE 7421.921617995082
MSE 85616476.3447491
RMSE: 9252.917180259914
```

In [28]:
```
import statsmodels.api as sm
```

In [29]:
```
X_train[['State_California', 'State_Florida']] = X_train[['State_California', 'Stat
X2 = sm.add_constant(X_train)
est = sm.OLS(y_train, X2)
est2 = est.fit()
print(est2.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 Profit   R-squared:                       0.957
Model:                            OLS   Adj. R-squared:                  0.951
Method:                 Least Squares   F-statistic:                     151.4
Date:                Mon, 27 Apr 2026   Prob (F-statistic):           3.23e-22
Time:                        18:51:11   Log-Likelihood:                -420.56
No. Observations:                  40   AIC:                             853.1
Df Residuals:                      34   BIC:                             863.3
Df Model:                           5
Covariance Type:            nonrobust
=====================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------
const             4.717e+04   8093.202      5.828      0.000    3.07e+04    6.36e+04
R&D Spend            0.8201      0.053     15.599      0.000       0.713       0.927
Administration      -0.0091      0.063     -0.144      0.887      -0.138       0.120
Marketing Spend      0.0230      0.019      1.191      0.242      -0.016       0.062
State_California -1352.0119   3810.480     -0.355      0.725   -9095.839    6391.815
State_Florida      -74.7329   3764.195     -0.020      0.984   -7724.498    7575.032
==============================================================================
Omnibus:                       12.497   Durbin-Watson:                   1.782
Prob(Omnibus):                  0.002   Jarque-Bera (JB):               14.566
Skew:                          -0.982   Prob(JB):                     0.000687
Kurtosis:                       5.210   Cond. No.                     1.53e+06
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spe
cified.
[2] The condition number is large, 1.53e+06. This might indicate that there are
strong multicollinearity or other numerical problems.
```

In [30]:
```
lm2 = LinearRegression()
lm2.fit(X_train[['R&D Spend']],y_train)
```

Out[30]:
```
▾ LinearRegression
▶ Parameters
```

In [31]:
```
predictions2 = lm2.predict(X_test[['R&D Spend']])
np.corrcoef(y_test,predictions2)[0][1]
```

Out[31]:
```
np.float64(0.9381960295948546)
```

In [32]:
```
#Correlacion del modelo anterior
np.corrcoef(y_test,predictions)[0][1]
```

Out[32]:
```
np.float64(0.9501983359183575)
```

In [33]:
```
lm2.coef_
```

Out[33]:
```
array([0.8657168])
```